

Ambareesh Ramakrishnan

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Passionate software engineer with expertise in Generative AI, full-stack, and data engineering. Skilled at building scalable solutions across various domains using modern technologies and cloud platforms.

EDUCATION

Master's of Science in Computer Science with AI Concentration <i>Oregon State University (CGPA : 4/4)</i>	Sep 2023 - March 2025 (<i>Expected</i>)
Bachelor of Technology in Information Technology <i>CEG, Anna University (CGPA : 8.4/10)</i>	Aug 2015 - Apr 2019

WORK EXPERIENCE

Oregon State University <i>Research Assistant</i>	Sep 2023 - Present <i>Oregon, USA</i>
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- **AI Cortex - CI/CD Pipeline Optimizer:** Built a hybrid RAG system for GitHub Actions using semantic search and dense retrieval. Implemented agentic workflows with custom function tools for real-time analysis, achieving 92% accuracy in identifying pipeline inefficiencies across 1000+ workflows (React, Node.js, OpenAI API, LangChain, GitHub API)
- **AID (Automated Inclusivity Detector):** Developed an AI-powered system to detect inclusivity issues in MOSIP's platform applications, integrating GenderMag data and multi-modal prompting for UI analysis. Achieved 78% accuracy in identifying cognitive-bias issues through RAG-based analysis. (Python, OpenAI, AWS Bedrock, ChromaDB, Redis)
- **Advanced Prompting Strategies :** Implemented specialized techniques including **graph-based prompting** for YAML serialization, **self-consistency prompting** for diverse bug detection, and **chain-of-thought reasoning** with few-shot learning. Improved overall suggestion accuracy from 45% to 86%.
- **Feedback-Driven Optimization:** Designed an automatic prompt refinement system based on user interactions and feedback metrics, reducing false positives by 32%. Contributing to research on creating LLM with goal maintenance loops.

VISA <i>Senior Software Engineer</i>	July 2019 - Aug 2023 <i>Bangalore, India</i>
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- **Data Governance Platform:** Architected and led a team of 5, enabling automated data validation with real-time notification and secure role-based access controls, and data archival/retrieval. Reduced data correction time by 70% across Visa HR products and saved \$250,000 annually in license costs. (Angular, .NET 6, Kotlin, SQL, MongoDB, PowerBI)
- **Integration Hub :** Served as tech lead for the India team, managing 10+ client integrations, optimizing performance, and implementing centralized logging and automated token authorization. (Microsoft BizTalk, .NET, Scripting, MSSQL).
- **AI Chatbot:** Contributed to the development of an HR chatbot proof-of-concept using Azure AI services (QnA Maker and Azure OpenAI) to automate responses to common HR queries. (.NET, Azure Storage, Angular, SQL)
- **Cube Performance Optimization:** Achieved a 150% increase in PowerBI dashboard performance by optimizing OLAP cubes, implementing efficient ETL pipelines, and designing snowflake schemas. (SSAS, DAX, M, SQL)
- **Application Management and Optimization:** Managed 5+ applications, streamlining processes and optimizing DevOps workflows by 20%. Conducted security assessments, resolving 35+ critical vulnerabilities. (Azure DevOps, VVMS)
- **Visa Web Application Framework:** Key contributor to VWAF, which significantly reduced development time of new applications. Developed modules for authentication, caching, encryption, CRUD scaffolding, DevOps, and UI components.

Sigaram Technologies <i>Software Development Intern</i>	April 2018 - June 2018 <i>Chennai, India</i>
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- **Order Insights and Reporting Module:** Developed a data analytics module aggregate orders and item data into KPIs, displayed via dashboards. Improving decision efficiency by 25% and contributed to 15% revenue growth. (React, Java, MySQL, Tableau)

PROJECTS

- **Deep Unlearning** - Conducted research comparing Amnesiac, Knowledge Transfer, and Impair Repair-based unlearning across CNN, ResNet-18, and ViT. Developed novel comparison metrics for privacy-compliant AI systems.
- **Automatic Code Documentation** - Created hybrid Transformer architecture combining code-trained encoder and language decoder for code summarization. Improved performance through end-to-end training on code docstring datasets.
- **Rumour Stance and Veracity Detector** - Built a multi-stage system, using Graph CRF for stance classification and stacked LSTM for veracity prediction (85% accuracy). Used Siamese Bi-LSTM and OpenKE for claim substantiation.
- **Question Duplicate Identification** - Developed Siamese LSTM network to identify semantically similar questions in large datasets. Improved information retrieval and query understanding for enhanced search engine performance.

SKILLS

Generative AI	Azure OpenAI, LangChain, Agentic Workflows, Neural Retrieval, RAG, Prompt Engineering
Languages/Frameworks	.NET, Angular, C#, C++, Flask, Kotlin, JavaScript, Python, PyTorch
Databases & Big Data	Azure Storage, MongoDB, Redis, SQL Server, SSAS, PySpark
Cloud/DevOps/Tools	AWS, Azure, Azure AI Search, Git, Power Automate, PowerBI, Selenium, Visual Studio

LEADERSHIP

- Led dev teams across multiple projects, mentoring junior engineers and new hires through regular code reviews and knowledge sharing sessions - recognized with **Above & Beyond awards** for team leadership and mentoring.
- Managed key **stakeholder relationships** for Integration Hub with SmartRecruiters team and Data Governance Platform with CloudPay, Fidelity Investments and Workday, to gather user requirements, and implement feedback.
- Served as **interim scrum master** for 3 projects, conducting daily standups, sprint planning, retrospectives, and demos while managing technical debt, UAT cycles, and release planning to ensure timely delivery.
- Presented complex **technical demos to senior leadership**, including AI and Deep Learning sessions for HR Leadership Team and CIT Chief Architect that highlighted potential AI-driven tools opportunities.

PATENTS, PUBLICATIONS AND ACHIEVEMENTS

- **Conference Publication** (ICSE - CSEE&T 2025) - “**Insights from the Frontline: GenAI Utilization Among Software Engineering Students**” - (Acceptance Rate - 12%).
- **Patent** PCT/US2023/075209 - “**Secure Authentication Using Software Application**” (Pending) – Filed Dec 19, 2023.
- **Patent** PCT/US2024/042106 - “**Detection Of Cross-Side Scripting Attacks Using Deep Learning Models**” (Pending) - Filed Sep 17, 2024.
- **Defensive Publication** 5677 - “*Secure Buy Now Pay Later (BNPL) for B2B*” using federated blockchain and Deep Learning methods.
- **Defensive Publication** 6071 “*Recommendation of Distribution Channels for Merchants*” using Machine Learning techniques and VisaNet data.
- **Trade Secrets** - Received 5+ **Technical Innovation Awards** for contributing to AI Referral, Automatic Anomaly Detection, and Fleet price prediction, recognized as **intellectual properties** by the company.
- **Bachelor’s Thesis** - Published a paper on “**Rumor Stance and Veracity Determination**” using Siamese Bi-LSTM and Knowledge Graph, achieving 87% accuracy on the Stanford NLI dataset (Ranked top 1% in SemEval).
- **Certifications** - **Deep Learning** and **Data Engineering** from Udacity, and **NLP Specialization** from Coursera.